

Nicholas Ketz PhD.

Applied Research Scientist: Artificial Intelligence, Bio-informatics, Mathematical Models, Human Studies

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Summary

I am an applied scientist developing machine learning solutions to advanced research problems. I have more than 10 years of experience in both academic and industrial research labs. I have experience individually contributing, leading small teams and coordinated across multiple departments and organizations throughout the entire AI product life cycle.

Select Experience

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|--|-------------------|
| Large Language Modeling | 03/2022 - Present |
| • Building DNA foundation models and fine-tuning them for gene therapy and cross-species engineering applications in biotechnology | |
| Generative AI | 01/2020 - 01/2022 |
| • Created AI systems for generating video and images grounded in learned human concepts and semantic understanding | |
| Model Based Reinforcement Learning | 01/2020 - 01/2022 |
| • Developed Agent-based domain adaptation techniques for autonomous driving using model-based reinforcement learning | |
| Lifelong Deep Learning | 09/2018 - 01/2021 |
| • Develop new methods for performance preservation in sequential multitask and multi-modal learning in deep neural networks | |
| Closed-loop Neural Stimulation | 09/2016 - 06/2018 |
| • Developed devices and algorithms for neural stimulation in human subjects to autonomously enhance cognitive performance | |

Education and Employment

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|--|-------------------|
| Principle AI/ML Scientist <i>Colossal Biosciences/Form Bio</i> | 02/2022 - Present |
| • Leading AI/ML research in cross-discipline team on the full AI product life cycle supporting various biotech clients | |
| Research Scientist: Information and Systems Sciences <i>HRL Laboratories</i> | 09/2016 - 01/2022 |
| • Basic AI research funded by Department of Defense and product development for aerospace and automotive industries | |
| PhD: Computational Cognitive Neuroscience <i>University Colorado, Boulder</i> | 09/2010 - 09/2016 |
| • Development of biologically inspired neural networks informed by neuro-imaging experiments of cognitive function | |
| Research Assistant: Davachi Human Memory Lab <i>New York University</i> | 09/2007 - 07/2010 |
| • Supported graduate level research for neuro-imaging of human long-term memory systems | |
| BA: Physics, minor: Psychology <i>University Minnesota, Twin Cities</i> | 09/2003 - 06/2007 |
| • Studied physics with a minor in psychology, with research experience in particle physics and cognitive-neuroscience | |

Relevant Skills

- **Technical:** Deep Learning (CNN, RNN, VAE, Transformers, Multi-Modal, RL), Machine Learning (Optimization, Probabilistic Inference, Unsupervised Learning), Statistics (Parametric, Monte Carlo, Bayesian, Time-series), Programming (Python, PyTorch, TensorFlow, Scikit-learn, Git, Docker, R, CUDA, GCP, MLOps)
- **AI Lifecycle:** Full-cycle AI development (proof-of-concept to production), cross-team product development, system design/orchestration with containerization, CI/CD pipelines with unit testing and quality assurance
- **Professional:** Team leadership, cross-departmental collaboration (engineering, product, sales), client/stakeholder management, project coordination across multiple research domains and organizations

Select Publications/Patents

- | | |
|--|-------------------|
| System and method for semantically grounded video generation | US Patent, 2024 |
| • Generative video AI system prompted by human interpretable neuro-symbolic concepts | |
| Model Interpretability Methods to Predict Gene Therapy Manufacturing Failures | White Paper, 2022 |
| • Extracting learned motifs from DNA sequence models predictive of gene therapy manufacturing failures | |
| Independent component selection for automated signal artifact removal | US Patent, 2020 |
| • Scalable solution for removal of nuisance components in time-series data | |
| Sliced cramer synaptic consolidation for preserving deeply learned representations | ICLR, 2019 |
| • Framework for overcoming neural network forgetting and improving transfer in sequential multitask learning | |